

Journal example

General Description:

Elaborate in general about the items included in the journal. Brief overview of the weekly logs, the understanding of current technology used in the subject.

Intent:

Elaborate in general about the study objectives (what you want to achieve in the subject), the learnings materials (describe your experience in using the subject learning materials), and the teaching facilities (your experience in the current computer labs or remotely).

The following are example of week 2

Week 2

Lecture 2 Installation and Configuration (Disk Management)

Brief description and elaboration about the week's lecture

Table #number [Setup and Configuration used in this lecture]

Diagram #number [Illustrate the concepts, entities, relationships]

Lessons #number [Outline the learned lessons and objectives]

Lab 2d Disk Partitioning

There are three steps in the process of making and using filesystems:

1. Divide the hard drive into partitions.
2. Format each partition with a filesystem.
3. Mount the formatted partitions onto the directory tree of an existing filesystem.

Task 1: To divide the hard drive into portions:

Options1: use fdisk:

Option2: GNU Parted

From man page:

parted is a program to manipulate disk partitions. It supports multiple partition table formats, including MS-DOS and GPT. It is useful for creating space for new operating systems, reorganising disk usage, and copying data to new hard disks.

GNU Parted program is a tool for creating and resizing partitions on a hard drive. GNU Parted will non-destructively resize a partition as well as the filesystem on top of it.

The `parted` program can be used in two ways: *command line* mode and *interactive* mode. When using `parted` in either mode, a device must be specified. To do this, the following syntax must be used:

```
parted DEVICE
```

The `DEVICE` argument is used to specify the hard drive to modify, for instance, `/dev/sdb`. If this argument is not used, a default device will be chosen for you, or an error message may be displayed. To use `parted` in command line mode, the `DEVICE` argument must be followed by options to create or modify a partition. To see a list of options available for `parted`, use the `--help` option:

```
root@localhost:~# parted --help
Usage: parted [OPTION]... [DEVICE [COMMAND [PARAMETERS]...]...]
Apply COMMANDs with PARAMETERS to DEVICE.  If no COMMAND(s) are given, run in
interactive mode.

OPTIONs:
  -h, --help                displays this help message
  -l, --list                 lists partition layout on all block devices
  -m, --machine              displays machine parseable output
  -s, --script               never prompts for user intervention
  -v, --version              displays the version
  -a, --align=[none|cyl|min|opt] alignment for new partitions

COMMANDs:
  align-check TYPE N        check partition N for TYPE(min|opt)
                             alignment
  help [COMMAND]            print general help, or help on
                             COMMAND
  mklabel,mktable LABEL-TYPE create a new disklabel (partition
                             table)
  mkpart PART-TYPE [FS-TYPE] START END make a partition
  name NUMBER NAME          name partition NUMBER as NAME
  print [devices|free|list,all|NUMBER] display the partition table,
```

```

    available devices, free space, all found partitions, or a particular
partition
quit                                exit program
rescue START END                    rescue a lost partition near START
    and END
resizepart NUMBER END               resize partition NUMBER
Output Omitted...

```

To view any existing partition information with the `parted` command, use the following:

```

root@localhost:~# parted /dev/sda print
Model: VMware,  VMware Virtual S (scsi)
Disk /dev/sda: 21.5GB
Sector size (logical/physical): 512B/512B
Partition Table:msdos
Disk Flags:

```

Number	Start	End	Size	Type	File system	Flags
1	1049KB	1075MB	1074MB	primary	ext4	boot
2	1075MB	21.5GB	20.4GB	primary		lvm

Options3: use blivet-gui software packages:

Task 2: Format each partition with a filesystem

The `mkfs` command can be used to create a filesystem. To make a vfat type filesystem, which is compatible with multiple operating systems, including Microsoft Windows, execute the command like the following:

```

root@localhost:~# mkfs -t vfat /dev/sdb1

```

Task3: Mount the formatted partitions onto the directory tree of an existing filesystem.

Mounting of partitions and checking on existing mounts is accomplished with the `mount` command. When called with no arguments, the `mount` command shows the currently mounted devices. This can be performed by regular users, not just the root user.

The output of the command may appear like the following:

```

sysadmin@localhost:~$ mount

```

```
/dev/sda2 on / type ext4 (rw)
tmpfs on /dev/shm type tmpfs (rw)
/dev/sda1 on /boot type ext4 (rw)
/dev/sda5 on /home type ext4 (rw)
none on /proc/sys/fs/binfmt_misc type binfmt_misc (rw)
sunrpc on /var/lib/nfs/rpc_pipefs type rpc_pipefs
```

The general format of this output is:

```
DESCRIPTION OF SOURCE on MOUNT POINT type FILESYSTEM TYPE (OPTIONS)
```

In order to properly mount a filesystem, at least two parameters must be provided to the `mount` command: the filesystem identifier and the directory path name to mount it on. If no other options are needed, the `mount` command will mount the filesystem on the specified directory mount point.

Note

Unless set up in the `/etc/fstab` file beforehand, mounting requires root user access.

```
root@localhost:~# mount /dev/sdb1 /mnt
```

To unmount a filesystem, use the `umount` command with either the filesystem or the directory given as an argument. For example, to unmount the `/mnt` directory that was previously mounted, execute the following command:

```
root@localhost:~# umount /mnt
```

Because the `/dev/sdb1` filesystem was mounted under the `/mnt` directory, the following command would have also worked:

```
root@localhost:~# umount /dev/sdb1
```

There are a few instances where the `umount` command will fail to unmount a filesystem:

- There is a file open that is located within the filesystem. The term *open* meaning *in use by a program*. For example, if the file was being edited by a text editor like the `vi` editor, it would be considered to be open.
- There is an executable file running that is located within the filesystem.
- A user has a directory within the filesystem as their current working directory.

Lab finding:

Mounting a filesystem on a directory that already contains files will render those files temporarily inaccessible.

Once a filesystem is mounted on a directory, only the contents of that filesystem will be visible. What was previously visible in a directory is not gone, but will become visible only after the new filesystem is unmounted.

Expansion

Manually mounting filesystems with the `mount` command results in a non-persistent mount. If the system is rebooted, then the filesystem must be mounted again to be accessed. The same is true for activating swap devices with the `swapon` command.

The `/etc/fstab` file is used to configure what filesystems will be mounted automatically at boot. It can also be used to activate swap devices automatically. Root privileges are required to make changes to the `/etc/fstab` file.

A system with three filesystem partitions and one swap partition might have an `/etc/fstab` file that looks like the following:

```
# /etc/fstab
# Created by anaconda on Fri Jan 17 10:31:51 2014
#
# Accessible filesystems, by reference, are maintained under '/dev/disk'
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info
#
/dev/sda1 /      ext4    defaults    1 1
/dev/sda2 /boot  ext4    defaults    1 2
/dev/sda5 /home  ext4    defaults    1 2
```

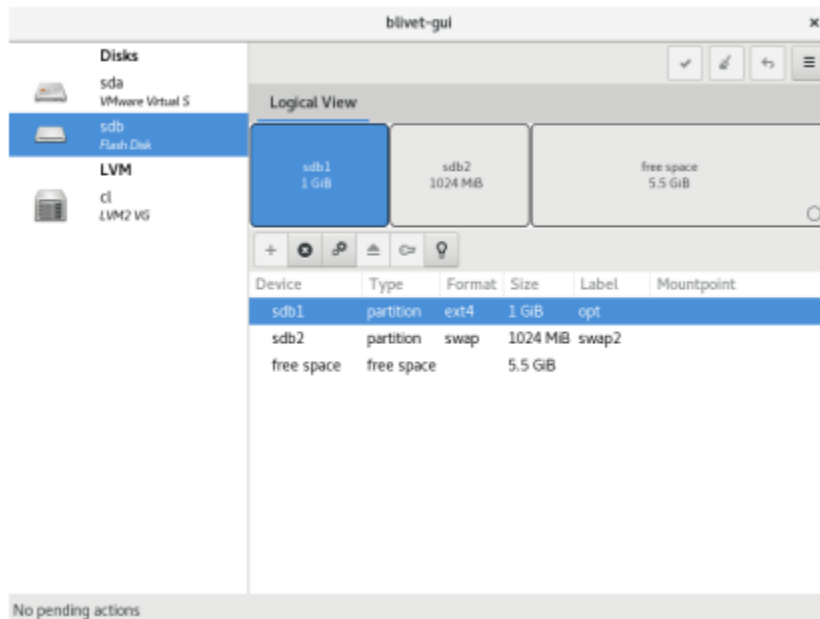
The first four columns in the `/etc/fstab` file map directly to the parameters used to mount a filesystem manually with the `mount` command.

Creating a Swap Partition

Swap space (also called virtual memory) is hard drive space that can be used by the kernel and memory management routines to store data that is normally stored in RAM. When RAM starts to become full, the kernel will take some of this data and swap it to the hard drive. At a later time, as it's needed, the data will be swapped back to RAM. typically the size of the swap space is at least equal to the size of RAM.

The steps to creating a swap partition are:

1. Create a partition: run blivet-gui to create partition



2. Convert the partition to swap space with the `mkswap` command.

```
root@localhost:~# mkswap /dev/sdb2
Setting up swapspace version1, size = 102396 KiB
no label, UUID=59aaf06e-7109-471f-88a5-e81dd7c82d76
```

3. Enable the partition as current swap space with the `swapon` command:

```
root@localhost:~# swapon /dev/sda2
```

The `-s` option to the `swapon` command will display currently used swap space:

```
root@localhost:~# swapon -s
```

Filename	Type	Size	Used	Priority
/devdm-1	partition	1015800	0	-1
/dev/sda2	partition	102392	0	-2

Note

The `swapon` command only temporarily enables a swap partition. If the system is rebooted, the swap partition will still exist, but it will not be enabled as swap space. The administrator needs to either execute the `swapon` command again or make use of a mounting feature that is demonstrated later in the course.

Reflection:

Elaborate thoroughly about your experience in the weeks for each part of the journal. Discuss in detail your learning experience in the class (On-Campus or Online). Discuss in detail your opinion about the learning environment and your overall opinion about the teaching style.

Discuss in detail the challenges you encountered and the problem-solving techniques and research you utilized to overcome the challenges presented by the subject and learning process.

Discuss in detail the challenges that remained unsolved and outline what you could have done better to resolve these issues.

Referencing:

Provide a bibliography section using Harvard reference style. Ensure that the references used in your research to acquire the knowledge and skills are properly indexed and sorted by alphabetical order. Ensure that the references are properly cited in the Reflection section.